Video Transcript

Video 2: Using Pandas to Read Big Data (02:50)

Now, we know from manipulating files, it's really nice if we can read into a Pandas data frame. So, let's take a look. We know we can read CSV files in Panda, so let's load up. We'll import Pandas as we normally do. We'll put our timing in here. Now, I happen to know the data types that the columns are using in the file, and I'm putting those in so it'll actually read it faster. So, we're going to read the file in as a CSV file, but there's perhaps a better way of handling these files.

And, Parquet is one. So, once we've got it read in, I'm going to write it out to Parquet. Maybe I'll call it my million lines. Let's write it out. So, example02, and we'll see how long this takes. I'm going to stop the video just to show you that we're using 3.38 gigabytes of memory at the moment here in our Python code, so it's quite voluminous. This is a four gigabyte, it went up to five quite suddenly, and we finished, excellent. So, we read in 101 million lines, and we did that in 187 seconds.

So, Pandas is not fast but it can read 100 million lines. That'll give you some idea of the size file. And now we wrote out our file, and this case we chose Parquet format and that's a better format for reading files in. Okay so, I wrote the file out for you in Parquet format, it's called riid_train.parquet. Why don't you read that back in? Tell me how long it takes to read then, write it out in a format called Feather. So, instead of Parquet here so, just replace it with Feather format and let me know how long that takes to read back in. Okay so, that's your exercise.

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